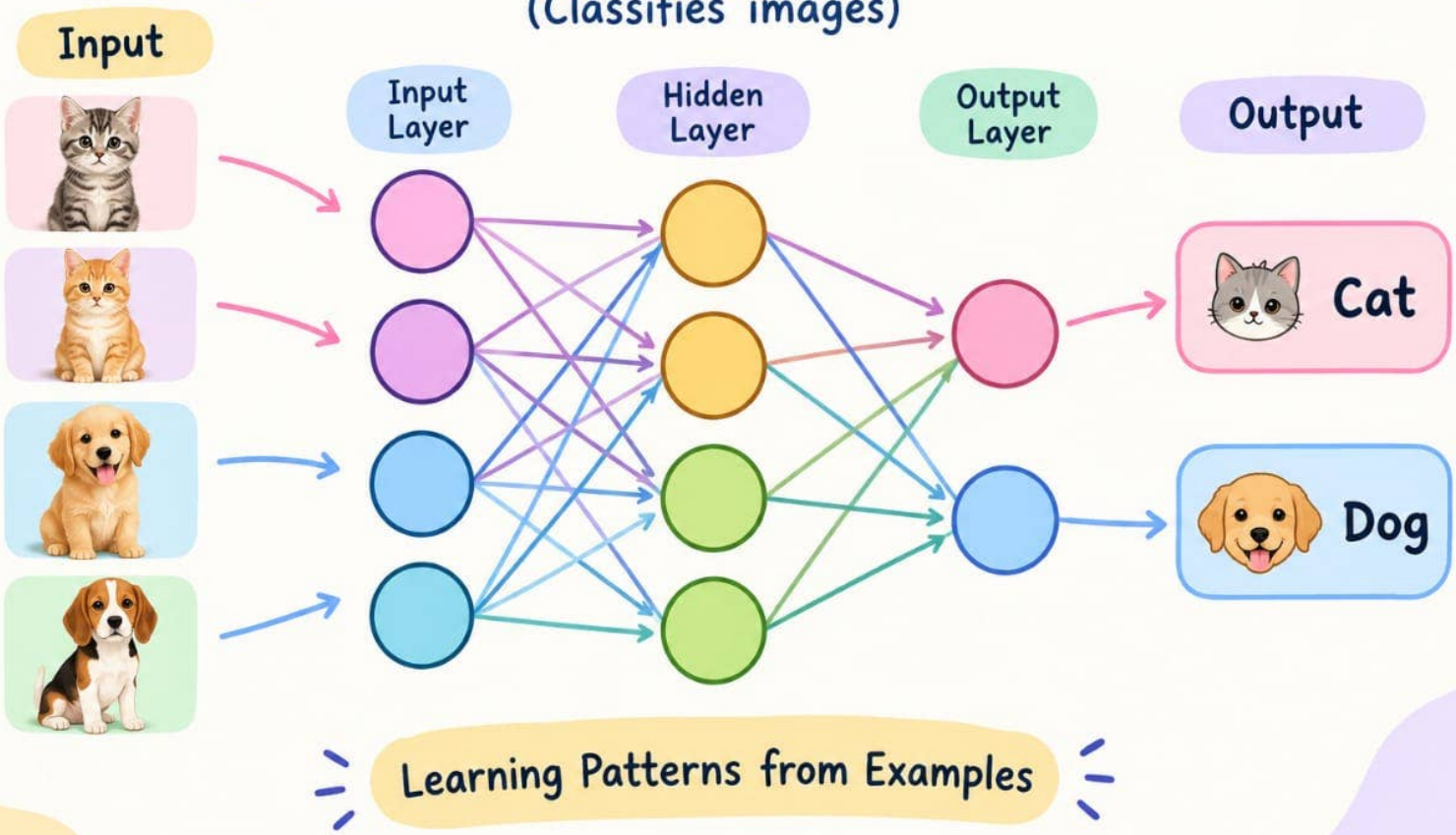
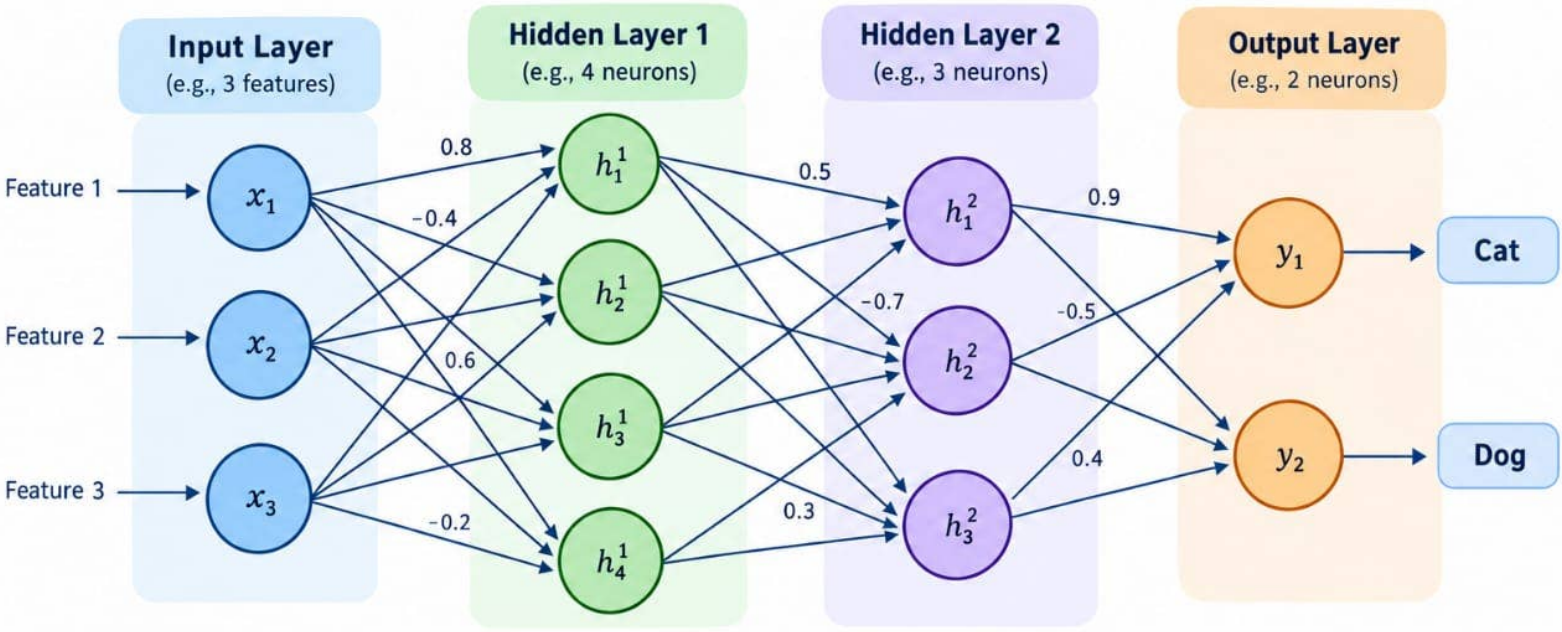
Artificial Neural Network

(Classifies images)



Learning Patterns from Examples

Architecture of an Artificial Neural Network



= Neuron (node)

= Connection (with weight)

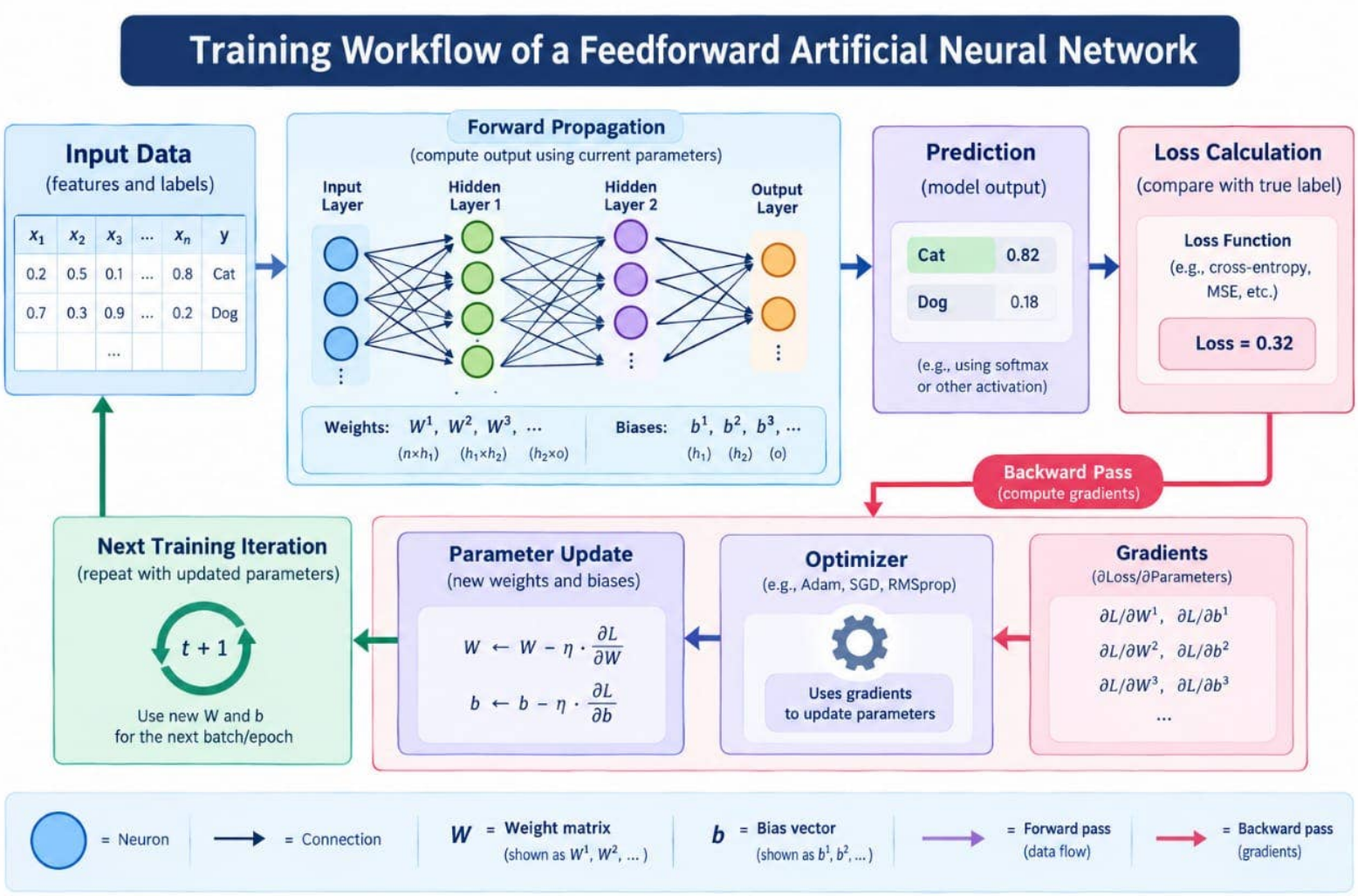
0.8 = Weight (example value)

How it works:

1. Input features enter the network.
2. Each neuron multiplies inputs by weights, sums them and applies an activation function.
3. The output layer gives the final prediction.

Note:
This is just one example of a feedforward neural network. Other networks may have different numbers of layers, neurons or connection patterns.

Training Workflow of a Feedforward Artificial Neural Network



= Neuron | → = Connection | W = Weight matrix (shown as W^1, W^2, ...) | b = Bias vector (shown as b^1, b^2, ...) | → = Forward pass (data flow) | → = Backward pass (gradients)